CORA: RANDOMIZED DECODING FOR QUANTIZED UPDATE INVERSION AND ITS RECONSTRUCTION LIMITS ACROSS BIT WIDTHS

Anonymous Author(s)

Anonymous Affiliation

ABSTRACT

Quantizing a model update leaves multiple numerical updates compatible with the transmitted observation. We study CoRA (Consistency-Ranked restart Attack), which explores this ambiguity through randomized decoding, source separation, and feature inversion. The sender quantizes deterministically; randomness enters only when the attacker samples admissible values within the observed quantization cells. A public score combining feature agreement and total variation ranks the resulting reconstructions, and selected candidates receive further optimization. We evaluate this procedure on VGG16/ImageNet local updates using signed per-tensor quantization from 8 to 4 bits. As precision decreases, the selected reconstruction's LPIPS increases from 0.5558 to 0.8085 and PSNR decreases from 12.0568 to 11.5028 dB. Reconstructions at 8 bits retain visible structure in some examples, whereas the 4-bit outputs exhibit substantial loss of object and scene detail. Relative to naive dequantization, the benefit of randomized decoding varies with bit width and metric. These results characterize both the behavior and the limits of CoRA under the evaluated reconstruction budget.

***Index Terms*—** Update inversion, randomized decoding, quantization, source separation, reconstruction quality.

1. INTRODUCTION

Model updates can reveal information about the data used to compute them. Deep Leakage from Gradients and Inverting Gradients reconstruct training inputs through gradient matching [1, 2]. The Cocktail Party Attack (CPA) instead uses independent component analysis to separate the inputs mixed in fully connected gradients and updates [3]. In a compressed communication setting, however, an attacker receives a quantized representation rather than the original numerical update. The reconstruction procedure must therefore choose how to decode it.

Naive dequantization assigns every received code its cell center. This is a natural deterministic estimate, but it represents only one numerical completion of the observation. A randomized decoder can generate other admissible completions and pass them through the same reconstruction pipeline. The important empirical questions are whether candidate search helps recover image structure and how the resulting quality changes when the quantization cells become wider.

We address these questions with CoRA, a randomized decoding and candidate-selection procedure. It samples values inside the observed cells, separates source features from each sampled update, and inverts those features into images. A score computable from the public model and the recovered features determines which candidates receive additional optimization. The output is one selected reconstruction; source or image averaging is not part of the evaluated method.

The paper makes three contributions. First, it specifies a compression-compatible randomized decoder and its integration with a CPA-based source-separation and feature-inversion pipeline. Second, it defines a public candidate-ranking procedure and reports its actual computational budget. Third, it characterizes reconstruction quality across an 8/7/6/5/4-bit per-tensor sweep, using paired naive-decoding controls, image metrics, and qualitative reconstructions. Our emphasis is the observed dependence on bit width, including settings where randomized search provides little or no improvement.

2. OBSERVATION MODEL

2.1. Client update and attacker knowledge

We consider an exposed single-client local update in a federated learning round. The client starts from public weights and performs full-batch SGD. The convolutional feature extractor is frozen, while the fully connected layers are trained. The attacker knows the starting model, batch size, and local training configuration, and observes the quantized fully connected updates with their decoding metadata. The attack uses the first fully connected weight matrix. Training images, labels, true features, and uncompressed updates are unavailable to the attack and its candidate selector.

Let $\mathbf{H} = [\phi(x_1), \ldots, \phi(x_B)]^\top$ contain the frozen features entering the first fully connected layer. If the gradient at local step t is $\mathbf{A}_t\mathbf{H}$, with loss averaging included in $\mathbf{A}_t$, then the local update is

$$\mathbf{U} = \mathbf{W}^{(T)} - \mathbf{W}^{(0)} = -\eta \left(\sum_{t=0}^{T-1} \mathbf{A}_t \right) \mathbf{H}. \qquad (1)$$

This expresses the multi-step update as a mixture of fixed input features and motivates CPA-based source separation. The sign convention matches the stored after-training-minus-initial update. Our observation model does not assume access to an individual update concealed by secure aggregation.

2.2. Signed per-tensor quantization

For each tensor, the deterministic quantizer uses

$$Q_b = 2^{b-1} - 1, \qquad s = \frac{\|\mathbf{U}\|_\infty}{Q_b}, \\ q_j = \mathrm{round}_{\mathrm{even}}(U_j/s). \tag{2}$$

Codes lie in $[-Q_b, Q_b]$, giving $2^b - 1$ numerical levels, and one FP32 scale s is stored per tensor. A zero tensor is represented by zero codes and unit scale. Arithmetic and scale storage follow the FP32 implementation. Throughout this paper, bit width refers to numerical code precision. The experiment artifacts store codes in int8 with additional scale and packet metadata; this sweep is not a measurement of packed communication size.

For positive s, the ideal interior cell for code q_j extends from $s(q_j - \frac{1}{2})$ to $s(q_j + \frac{1}{2})$. Ties-to-even rounding, finite precision, and the requirement to reproduce the tensor scale determine the actual admissible endpoints. We write $\mathcal{C}(q, s)$ for the set of FP32 tensors that reproduce the received codes and scale. Naive decoding uses $U_j^{\mathrm{ctr}} = sq_j$.

3. CORA: RANDOMIZED DECODING AND CANDIDATE RANKING

3.1. Sampling compatible numerical updates

CoRA generates candidate first-FC matrices $\widetilde{\mathbf{U}}^{(m)} \in \mathcal{C}(q, s)$. For non-anchor coordinates, it draws uniformly between the admissible cell endpoints and casts to FP32. An extremal coordinate is anchored to a scale-compatible maximum. The implementation checks that the resulting matrix reproduces both the codes and the stored scale under re-encoding. This is a bounded sampling rule, without a learned or calibrated prior over the original update.

Naive decoding and one randomized draw are different procedures. The former always returns the cell center; the latter chooses another compatible value. In the ideal interior-cell model, center decoding has coordinate error at most $s/2$, whereas a sampled completion can differ from the true value by almost s. Randomization therefore does not guarantee a more accurate numerical update. Its purpose here is to explore alternative inputs to the nonlinear reconstruction pipeline.

3.2. Source separation and feature inversion

Each decoded first-FC matrix is processed using FastICA initialization followed by gradient refinement of the source separation. Source rows are aligned to the center-decoded reference using Hungarian assignment on absolute cosine similarity. Alignment changes permutation and sign only; it neither rescales source norms nor averages sources across members. The absolute recovered sources are used as feature-inversion targets, denoted $z_{m,i} = |\hat{h}_{m,i}|$.

For each candidate, feature inversion optimizes images $\hat{\mathbf{X}}_m$ to match these targets through the public feature extractor. The implemented backend is a modified CPA-based pipeline: FastICA and gradient refinement produce the features, and feature inversion produces the images. The reported results should be interpreted for this backend.

3.3. Public screening and continuation

After an initial feature-inversion budget, CoRA computes

$$J_m = \frac{1}{B}\sum_{i=1}^{B}[1 - \cos(\phi(\hat{x}_{m,i}), z_{m,i})] + 0.1\,\mathrm{TV}_{\mathrm{norm}}(\hat{\mathbf{X}}_m). \tag{3}$$

The score is evaluated on a pixel-boxed clone in normalized model-input space. It uses only the public model, decoded sources, and current candidate images. Here, consistency refers to feature agreement, regularized by total variation; this score does not replay local training or compare a recompressed image-induced update with the packet.

In the evaluated configuration, $K = 8$ sampled candidates each receive 5,000 feature-inversion iterations. The three lowest-scoring candidates continue to a total of 25,000 iterations, preserving optimizer and scheduler states. The scheduler horizon is 25,000 from the first step, and ties use ascending member index. The candidate ranked first at screening remains the primary reported output after continuation. We do not choose among the finalists using private image metrics or average their reconstructions.

This yields the following sequence for a fixed observed packet:

1. Decode the center reference and generate K compatible samples.
2. Separate source features for each decoding and align the sources.
3. Run initial feature inversion and rank candidates by J_m.
4. Continue the retained candidates and return the originally rank-1 candidate at the final optimization budget.

The naive control uses center decoding and the same reconstruction backend, with 25,000 feature-inversion iterations.

4. EXPERIMENTAL EVALUATION

4.1. Settings and evaluation protocol

We evaluate pretrained VGG16 on a fixed ImageNet batch, holding the victim data and local-training configuration constant across bit widths. The client uses $B = 16$, $T = 10$, and SGD learning rate 10^{-3}, with zero momentum, zero weight decay, and disabled dropout. Only fully connected parameters are trained. The quantizer uses one scalar scale per parameter tensor in every condition.

FastICA uses an iteration cap of 200 and tolerance 10^{-6}, followed by 4,000 refinement iterations at learning rate 10^{-3}. The source count is B. Randomized decoding uses seeds $1234 + 1000003m$ for $m = 0, \ldots, 7$. Feature inversion uses Adam with initial learning rate 0.1 and a cosine scheduler; its feature and total-variation weights are 1 and 0.1. These settings are held fixed across the bit sweep.

We report mean PSNR, SSIM, and VGG-based LPIPS [4]. Higher PSNR and SSIM, and lower LPIPS, indicate better image agreement. Private evaluation begins only after the public candidate selection has been fixed. Because source separation is permutation-ambiguous, the evaluator performs one pixel-MSE Hungarian assignment per candidate and reuses it for all three metrics and the displayed reconstructions. LPIPS uses FP32 RGB images mapped from $[0, 1]$ to $[-1, 1]$. All reconstructed images contribute to the reported means.

4.2. Combined compression results

Table 1 groups methods by dataset and presents compression settings as columns, with separate quantization and sparsification blocks for CIFAR-100 and ImageNet. The CIFAR-100 entries report the supplied baseline and random-selected results. The measured ImageNet blocks use different attack and evaluation protocols, specified below, so their absolute scores are not a controlled comparison of the two codecs.

In the ImageNet quantization block, selected CoRA LPIPS increases from 0.5558 at 8 bits through 0.6332, 0.6659, and 0.7490 to 0.8085 at 4 bits. PSNR decreases from 12.0568 to 11.5028 dB, while SSIM is not monotonic. Relative to naive decoding, CoRA improves SSIM and LPIPS at 8 bits with nearly unchanged PSNR, and all three metrics at 6 bits. The changes at 7 and 5 bits are mixed; at 4 bits, all three metrics worsen.

4.3. Sparsification: naive and random filling

The sparsification block uses a separate matched ImageNet study with frozen convolutional features and the same $B = 16$, $T = 10$ local-training configuration. The first-FC update retains its largest-magnitude entries at keep fractions of 50, 30, 20, and 10%; naive decoding fills the remaining entries with zero. These runs use the original CPA optimizer for 25,000 iterations and FIA for 25,000 iterations. Offline evaluation uses the saved source-cosine Hungarian assignment for all image metrics, rather than the pixel-MSE assignment used in the quantization block.

At 10% retention, a separate randomized procedure fills only discarded entries uniformly in $[-\tau, \tau]$, where τ is the smallest retained magnitude. Retained entries remain unchanged. Twenty candidates each receive CPA for 25,000 iterations and FIA for 5,000 iterations. An observable update-consistency score selects three candidates to continue FIA to 25,000 total iterations, then selects one final whole-batch output. The score replays the known local SGD procedure using public-model predicted labels, recompresses the update, and measures its relative ℓ_2 discrepancy from the received packet. This selector and backend differ from the quantization CoRA configuration.

Naive LPIPS increases from 0.4864 at 50% retention to 0.6323 at 10%. For these updates, 50% retention removes only already-zero coordinates. At 10%, random filling gives LPIPS 0.6300, PSNR 21.5104 dB, and SSIM 0.2610, versus

Table 1. Naive versus randomized decoding (PSNR↑, SSIM↑, LPIPS↓). Dashes denote unavailable results. Protocols differ between panels (Section 4.3).

(a) Quantization

Dataset	Method	Metric	8-bit	7-bit	6-bit	5-bit	4-bit
CIFAR-100	Baseline	PSNR	11.792	11.676	12.057	11.874	11.990
		SSIM	0.1200	0.1163	0.1203	0.1212	0.1285
		LPIPS	0.7550	0.7469	0.7442	0.7570	0.7372
	Random selected	PSNR	11.967	11.836	11.948	11.920	11.736
		SSIM	0.1090	0.1297	0.1250	0.1244	0.1207
		LPIPS	0.7279	0.7471	0.7429	0.7411	0.7486
ImageNet	Naive	PSNR	12.06	11.80	11.64	11.55	11.52
		SSIM	0.2924	0.2764	0.2596	0.2620	0.2921
		LPIPS	0.5594	0.6118	0.6961	0.7826	0.8071
	CoRA	PSNR	12.06	11.83	11.67	11.55	11.50
		SSIM	0.3070	0.2827	0.2690	0.2558	0.2576
		LPIPS	0.5558	0.6332	0.6659	0.7490	0.8085

(b) Sparsification: retained fraction

Dataset	Method	Metric	80%	50%	30%	20%	10%
CIFAR-100	Baseline	PSNR	11.994	12.020	11.948	11.793	11.757
		SSIM	0.1289	0.1304	0.1293	0.1266	0.1285
		LPIPS	0.7480	0.7528	0.7351	0.7467	0.7438
	Random selected	PSNR	11.994	11.911	11.845	11.812	11.855
		SSIM	0.1289	0.1306	0.1281	0.1145	0.1119
		LPIPS	0.7480	0.7424	0.7438	0.7432	0.7299
ImageNet	Naive	PSNR	—	22.29	22.14	21.87	21.53
		SSIM	—	0.3344	0.3141	0.2906	0.2631
		LPIPS	—	0.4864	0.5098	0.5536	0323
	Random fill	PSNR	—	—	—	—	21.51
		SSIM	—	—	—	—	0.2610
		LPIPS	—	—	—	—	0.6300

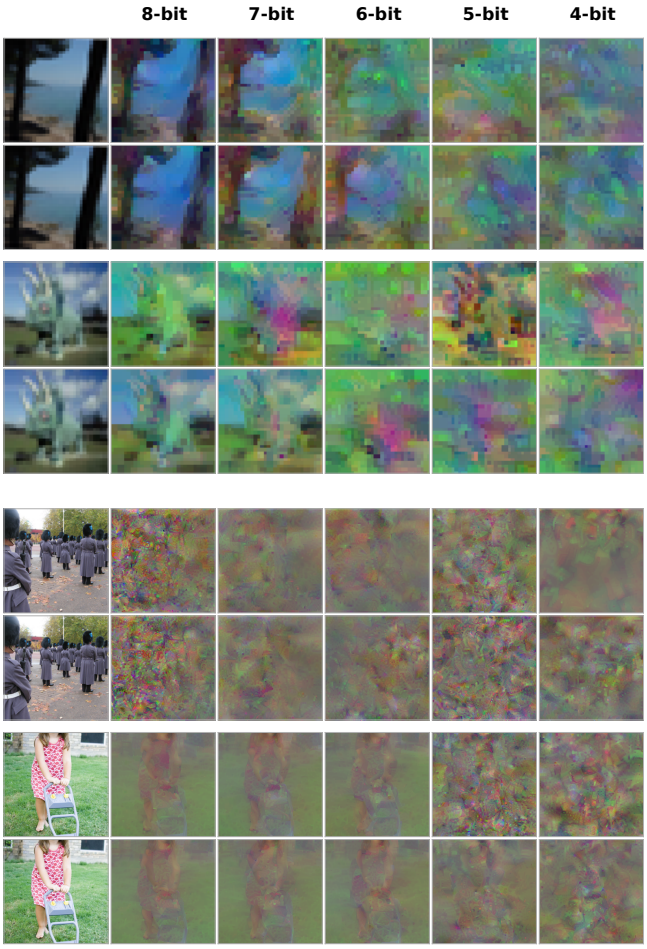


Fig. 1. Paired reconstructions at all five bit widths. Top four rows: CIFAR-100; bottom four: ImageNet. The first column shows originals. Within each two-row pair, Naive is above Random. CIFAR-100 examples are selected post hoc from 24 supplied examples with $R = 8$ outputs; ImageNet examples are fixed independently of quality, using publicly selected CoRA after 25,000 FIA iterations. Saved pairings are preserved, including pixel-MSE matching for ImageNet. These qualitative cases do not establish a full-set advantage.

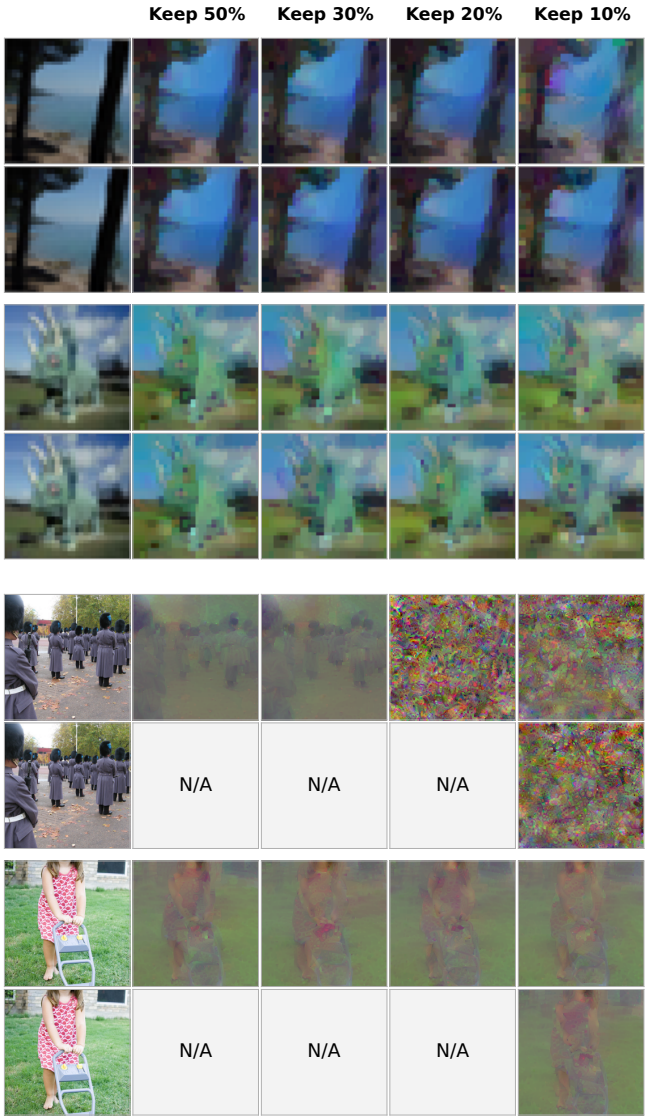


Fig. 2. Paired sparsification reconstructions at each retained fraction. Top four rows: CIFAR-100; bottom four: ImageNet. The first column shows originals; each two-row pair shows Naive above Random. CIFAR-100 uses the same selected examples as Fig. 1 with supplied $R = 8$ outputs. ImageNet uses fixed examples from the matched same80 study, with zero-fill Naive and publicly selected $R = 20$ Random. Stored pairing is preserved. N/A marks unavailable matched ImageNet Random results outside 10% retention. The protocols differ, so this is not a controlled cross-dataset comparison.

0.6323, 21.5305 dB, and 0.2631 for naive decoding. The change is small and mixed. Matched ImageNet random-fill results at the other retention levels are unreported and remain blank in the table and figure.

4.4. Individual draws and the selected candidate

At 4 bits, the first draw (member 00) has PSNR 11.5558 dB, SSIM 0.2614, and LPIPS 0.8055. It reached the final budget as the third-ranked candidate, not as a separately evaluated $K = 1$ procedure. The public selector chose member 06, whose image metrics are worse in this example. Thus public feature agreement is not equivalent to private image similarity. Table 1 retains the publicly selected candidate rather than replacing it with a more favorable private result.

4.5. Visual comparison with naive decoding

Figures 1 and 2 compare each randomized output directly with naive decoding of the same observation. The fixed ImageNet quantization examples include both a poorly recovered image and a person whose outline is visible at higher precision but deteriorates at lower precision. The sparse examples show the effect of reducing retention and the close visual agreement of the two methods at 10% retention. These examples supplement the full-set mean metrics; they do not define a universal reconstruction cutoff.

The CIFAR-100 rows in Fig. 1 use supplied examples 11 and 24 (one-based display indices), selected after viewing all 24 examples rather than fixed before evaluation. These cases illustrate differences in retained scene structure at lower precision; fine detail remains limited and the contrast varies with bit width. All five bit widths are shown for both methods. Whole image tiles and their saved pairing are retained without enhancement, averaging, or candidate reselection. This qualitative selection does not change the aggregate results or establish a general low-bit recovery advantage.

4.6. Computational budget and scope

The recorded randomized search performs $8 \times 5{,}000 + 3 \times 20{,}000 = 100{,}000$ feature-inversion iterations per bit width. The separate naive control adds 25,000 iterations, bringing the recorded total to 125,000. Source separation is performed for each of the eight samples and for the center reference. Thus final per-candidate iteration counts match, while total search budgets do not. The comparison measures the outputs of these procedures; it does not establish a compute-matched advantage from randomized decoding alone. For the sparse random-fill comparison, per-batch work is 500,000 CPA iterations, 160,000 FIA iterations, and 23 local-update replays, versus 25,000 CPA and 25,000 FIA iterations for naive decoding. Its total search budget is likewise unmatched.

The quantization sweep holds one victim batch and one set of candidate seeds fixed. It characterizes this setting, without estimating variability across independent batches or complete search runs. Comparisons to other datasets, quantization granularities, source-separation backends, or image-matching protocols require separate evidence.

5. DISCUSSION

Lower precision widens the numerical ambiguity available to the attacker, but the additional possible completions do not supply information about which one matches the private update. Equation 1 also imposes source structure that independent cell sampling does not explicitly enforce. Coarse completions can therefore provide poor inputs to source separation even when they reproduce the packet. This is a possible explanation of the observed degradation, not a mechanism isolated by the current sweep.

Candidate ranking presents a second limitation. The feature targets in Eq. 3 are themselves estimated from the compressed update. A reconstruction may fit those targets without resembling the private image, as the individual-candidate comparison illustrates. In addition, exact re-encoding of a sampled numerical update does not establish that local training on the final reconstructed images would reproduce the same packet. That image-induced update constraint was not evaluated in the quantization sweep; the separate sparse study uses update replay for candidate ranking.

The results consequently support a conditional account of reconstruction limits: with this sampler, selector, backend, and budget, 8-bit outputs retain structure in some examples, and lowering precision to 4 bits substantially degrades that structure. The intermediate-bit measurements describe the quality trajectory. They do not establish that any bit width protects against other attack procedures or larger search budgets.

6. RELATED WORK

Optimization-based gradient inversion reconstructs inputs by matching observed and synthetic gradients [1, 2]. CPA instead treats fully connected gradients as a blind source separation problem solved by ICA, enabling reconstruction at larger batch sizes [3]. Zhang et al. [5] survey gradient inversion attacks and defenses, including gradient protection strategies, and organize the main attack paradigms and reconstruction techniques.

Compression and other forms of gradient obfuscation have also been examined as defenses. Yue et al. [6] demonstrate that image information can remain recoverable after quantization, sparsification, or perturbation, and caution against treating privacy as a byproduct of gradient compression. Li et al. [7] introduce Generative Gradient Leakage, using a GAN prior learned from public images to reconstruct inputs under gradient degradation. Ding et al. [8] target sparsified gradients with Deep Leakage from Compressed Gradients (DLCG), which compresses synthetic gradients during reconstruction to account for the observed sparsity.

CoRA studies randomized decoding and public candidate selection without introducing a pretrained generative prior. The sender's quantization is deterministic; the attacker samples alternative numerical updates compatible with the observation. For quantization, candidates share a CPA-based reconstruction backend and are ranked by feature agreement and total variation. The separate sparsification study uses update-consistency ranking. Our evaluation compares these randomized procedures with naive decoding across bit widths and retention levels, reporting reconstruction quality and the actual search budgets. It characterizes the measured benefits and limits of candidate search under the evaluated protocols.

7. CONCLUSION

CoRA reconstructs quantized local updates by sampling compatible decodings and ranking the resulting image candidates using a public feature-based score. Its per-tensor VGG16/ImageNet bit sweep shows decreasing PSNR and worsening LPIPS from 8 to 4 bits, accompanied by a substantial loss of visual structure. Randomized decoding offers mixed improvements over naive dequantization and does not compensate for coarse quantization in the evaluated 4-bit setting. The measured quality curves and reconstructions describe where the current procedure retains image information and where its limitations become pronounced.

8. REFERENCES

[1] L. Zhu, Z. Liu, and S. Han, "Deep leakage from gradients," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[2] J. Geiping, H. Bauermeister, H. Dröge, and M. Moeller, "Inverting gradients—How easy is it to break privacy in federated learning?," in *Advances in Neural Information Processing Systems*, vol. 33, 2020.

[3] S. Kariyappa, C. Guo, K. Maeng, W. Xiong, G. E. Suh, M. K. Qureshi, and H.-H. S. Lee, "Cocktail Party Attack: Breaking aggregation-based privacy in federated learning using independent component analysis," in *Proc. International Conference on Machine Learning*, vol. 202, pp. 15884–15899, 2023.

[4] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition*, 2018.

[5] R. Zhang, S. Guo, J. Wang, X. Xie, and D. Tao, "A survey on gradient inversion: Attacks, defenses and future directions," in *Proc. International Joint Conference on Artificial Intelligence*, pp. 5678–5685, 2022.

[6] K. Yue, R. Jin, C.-W. Wong, D. Baron, and H. Dai, "Gradient obfuscation gives a false sense of security in federated learning," in *Proc. USENIX Security Symposium*, pp. 6381–6398, 2023.

[7] Z. Li, J. Zhang, L. Liu, and J. Liu, "Auditing privacy defenses in federated learning via generative gradient leakage," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10132–10142, 2022.

[8] X. Ding, Z. Liu, X. You, X. Li, and A. V. Vasilakos, "Improved gradient leakage attack against compressed gradients in federated learning," *Neurocomputing*, vol. 608, article 128349, 2024.